

I have never used the confidence band in practice, but there are several ways to calculate and plot it. Here, a calculation is proposed based on the t-distribution from `scipy.stats`, where `ip` is the part below (inferior) the model and `sp` the part above (superior) it. The data set is taken from Reference 7. A nice explanation about the confidence intervals is reported, for example, in Reference 2 pages 86-91.

```
from scipy import stats
n=w.rows
coef=polyfit(x,y,1)
yp=polyval(coef,x)
xx=linspace(min(x),max(x),1000)
dx=sum((x-mean(x))**2)
dy=sqrt(sum((y-yp)**2)/(n-2))
dd=((xx-mean(x))**2/dx)
t=stats.t.ppf(0.95,n-2)
ip=coef[1]+coef[0]*xx-(t*dy*sqrt(1/n+dd))
sp=coef[1]+coef[0]*xx+(t*dy*sqrt(1/n+dd))
plot(x,y,"o",markerfacecolor="lightgreen",zorder=3)
plot(x,yp,"k-",zorder=3)
fill_between(xx,ip,sp,where=sp>=ip,edgecolor="b",facecolor="b",
,alpha=0.2,interpolate=True)
```

The StatsModels package

Another way to build a regression model is by using the StatsModels package in combination with the Pandas package. An example is shown in the following code. The values for `x` and `y` are read from an `xls` file, then the weight is defined as $1/x^2$. Using the Pandas package, a `DataFrame` is defined for the couple `x`, `y` and a `Series` for the weights. Two types of regression are then calculated: OLS (Ordinary Least Squares, the one previously called the linear fit simple) and WLS (Weighted Least Squares). Last, both models are plotted with a simple `plot`. More information can be printed for slope `ols_fit.params[1]`, intercept `ols_fit.params[0]`, r-squared `ols_fit.rsquared`, a little report `ols_fit.summary()` or the equivalent for the weighted model using 'wls' instead of 'ols'. Two examples are shown in Figures 7 and 8. Further information, such as residuals and Cook's distance, can be printed or plotted using `resid`, `resid_pearson` and `get_influence()` respectively. A nice and large collection of examples is presented in Reference 8.

```
import pandas as p
import xlrd
from statsmodels.formula.api import ols,wls
w=xlrd.open_workbook("pk.xls").sheet_by_name("Sheet1")
x,y=w.col_values(0),w.col_values(1)
weight=1/array(x)**2
df=p.DataFrame({"Conc":x,"Resp":y})
w=p.Series(weight)
ols_fit=ols("Resp~Conc",data=df).fit()
```

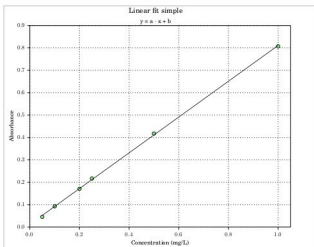


Figure 2: Python and LaTeX

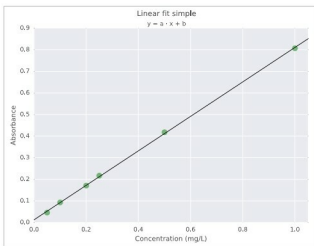


Figure 3: Pandas and Seaborn

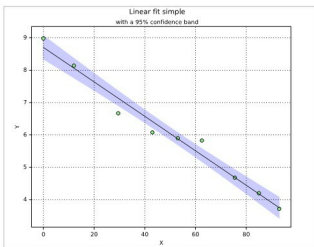


Figure 4: Confidence band

```
wls_fit=wls("Resp~Conc",data=df,weights=w).fit()
figure(0)
plot(x,y,"o",markerfacecolor="lightgreen",zorder=3)
```